

Enhancing Supply Chain Management Using Blockchain Technology

Keshav Gupta

Department of Computer Science
and Technology,
Delhi Technological University
Delhi, India
keshavgupta1999@gmail.com

Lakshay Goel

Department of Computer Science
and Technology,
Delhi Technological University
Delhi, India
lakshaygoel12@gmail.com

Manan Gupta

Department of Computer Science
and Technology,
Delhi Technological University
Delhi, India
manan8200@gmail.com

Nipun Bansal

Department of Computer Science
and Technology,
Delhi Technological University
Delhi, India
nipunbansal@dtu.ac.in

Abstract— A supply chain is a network of businesses and distribution choices that conducts the tasks of acquiring inputs, transforming these resources into variable and completed outputs, and distributing these manufactured goods to clients. The network is made up of many jobs at varying tiers of operation & distribution alternatives that carry out the work of procurement. It is necessary function in commercial and industrial sectors; nevertheless, the intricacy of the system differs significantly between industries. Blockchain is revolutionizing the modern landscape by providing a fresh viewpoint on access control, resilience, and overall performance. It provides a safe means of exchanging any type of item, service, or trade. Block chain technology would allow for further flexible production networks, quicker breakthroughs, and a somewhat intimate consumer interaction market. Such an article provides an outline of distributed ledger-tech as well as its possibilities for improving logistics operations.

Keywords—Blockchain, Supply Chain, Decentralized Distributed Ledger, Security, Global Value Chains

I. INTRODUCTION

The Blockchain model is a game-changing technology for organizations, with applications in cross-border payments, trade agreements, and repatriation. It could open up more possibilities across a variety of industries, including commerce, consumer packaged goods, medicine, bioscience, and entertainment. Blockchain, despite being a new technology, is fast evolving. Apart from private firms, some government agencies have begun to employ blockchain for administrative purposes on a modest scale, such as land transactions.

Traditionally, distribution channels in a supply chain are sequential in character, with a distinct flow of concept, strategy, purchase, manufacture, and dispatch. Logistics management is the discipline of coordinating the movement of foreign goods, infrastructure, data, and commodities from feedstocks to components suppliers to manufacturers to wholesalers to retailers to consumers. This path involves order advancement, contract receiving, data collection, and the accurate and faster delivery of the service. Traditionally, logistics managers sought for stability, capacity, speed, and clarity while attempting to enhance outcomes across a range of objectives such as overall spending, customer support, quality,

and aid for creativity. These traditional aims are unlikely to alter, though with technology and automation reinforced with digitalization methodologies, decision makers will be capable of reaching better levels of achievement.

Procurement professionals may assist in the establishment of new income streams by offering new and better economic opportunities and assisting the development of intelligent goods. These expansions will bring income to the supply chain's already broad set of objectives. The purpose of almost any supply chain is to transport resources, completed items, investment, and many other commodities from one location to another, and also to produce finished commodities. Rapid strong and innovative improvements, like increased processing capacity and substantially reduced prices, have had a significant influence on the conventional supply chain, along with a drop in transfer rates and an increase in development connected to the manufacturing method altogether.

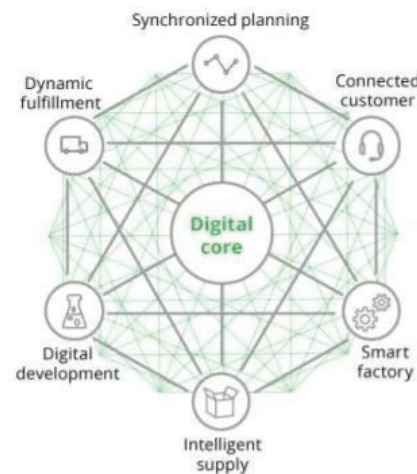


Fig. 1. A Supply Chain integrated using Blockchain

Blockchain is a growing list of blocks that are linked with each other as well as are constantly expanding as transactions are stored on the nodes. This technology employs a decentralized method, allowing data to be transmitted and each node of accessible data, also known as data, to have joint ownership. Blockchains store sets of encrypted transactions,

ensuring security, and are administered by community networking. Several advantages of a blockchain include confidentiality, privacy, and maintainability with no other influence from a third world party such as government, banks, company etc. Blockchain is highly resistant to data alteration due to its design. The year 2008 marked the beginning of the widespread adoption of blockchain technology, which was first envisaged in Nakamoto's studies. A new block is formed when a new transaction is transmitted by a user on the blockchain network. A block is used to store transactions on the blockchain, and all these blocks are transmitted to all of the network's connected systems. This transaction, which is contained within a block, is aired to all systems within the network. Every intermediate node has a copy of the whole blockchain, which aids in the authentication phase. When a block holding a user activity is broadcasted to all network participants, they check to see whether it has been interfered with in any way. If the confirmation is successful, the miners upload the block to their own blockchain copies.

In blockchain, the two most well-known algorithms in view of majority opinion are: Proof of Work (PoW) and Proof of Stake (PoS). In terms of its scope, the miners challenge with each other to find a right result to a computationally challenging job. The digital signature of the blockchain data and a varying data is calculated by the miners using their own processing capacity in PoW. The standard defines that the hash value should meet specific requirements. When a miner discovers a legitimate hashing, then it is credited with bitcoin in exchange for their computing strength. PoS was presented as a solution to the low efficiency of processing power. There has been no miners under PoS, hence there really are no miners. The machine's users are picked depending on a number of variables. One is "stake," which indicates the quantity of bitcoin that must be held up in advance. Whenever a user is selected, they must submit a block, that is then verified by the auditors. They will be rewarded if this component is legitimate. The stake is considered when any operator is proved to be fraudulent. The mechanism works by allowing any network device to validate the authenticity of a ledger in order to modify the blockchain with an entry. The blockchain, on either side, is utilized to accomplish the primary goal of blockchain: transferring data across users. The PKI generates a distinctive codename for each member in the system, verifying the identification of the users engaged. When a communication occurs, a transaction is formed to record the parties' addresses, signatures, and data. To store transaction data in a Merkle tree, a block is constructed using the hash value of the preceding block and a timestamp. It's worth noting that the chain's first block has no preceding block. Because of its structure, blockchain data is resistant to alteration. To account for varied use-cases, there are three main types of blockchain. These are the following: All participants have access to a public blockchain, which allows anyone can read data from it and even initiate new transactions to attach to it. As a result, even an attacker can access this form of blockchain.

The most common blockchains are public blockchains, which include cryptocurrencies such as Bitcoin. A consortium blockchain (also known as a semi-private blockchain) allows multiple parties to confirm blocks. The consensus algorithm in

a consortium blockchain is controlled by a large set of systems in the network. The data can still be read by participants, but not by the broader public. When numerous linked, but not mutually trusted, entities are working together or requiring data sharing, consortium blockchains are employed. Private: In a private blockchain, an entity or organization keeps the write rights, while the read authorization may be open to the public or to specific parties.

Private blockchains are frequently used in government and banking sectors where access control is required but traditional data storage methods are inefficient. In a private blockchain, an entity or organization keeps the write rights, while the read authorization may be open to the public or to specific parties.

Each of these blockchain types has its own set of benefits and drawbacks. While public blockchains are completely decentralized, the data they contain is vulnerable to hackers. When data exchange is required, consortium blockchains are effective. Its security, on the other hand, is contingent on the system and security mechanisms used outside of the model. Private blockchains eliminate the qualities of decentralization.

II. RELATED WORK

This study focused on how blockchain may improve supply chain visibility, although theoretical and practitioner use of blockchain technology is now constrained by our knowledge. To solve this issue, the UTAUT and the notion of technological innovation acceptance are used as a basic framework for procurement accountability in this study. A conceptual model is built, and the research concludes with smart contracts production chain considerations influenced by concept and previous research.[1]

In this research, a software gateway was developed and tested to communicate an Ethereum-like blockchain with the computer systems of businesses, allowing firms to gather relevant data with their associates with varying levels of clarity and to verify data authenticity, trustworthiness, and issues such as security over time via the blockchain, thereby ensuring accountability. To investigate the feasibility of using smart contracts in a logistics system, a simulator model was created to replicate inventory management and was coupled with the blockchain via the same computer adapter to do a scenario data study.[2]

This article explains how blockchain may be incorporated into distribution network design to produce a system that is dependable, accessible, genuine, and protected. To do this, they investigated the benefits of incorporating blockchain technology into the supply network as well as the problems found in a blockchain-based inventory control environment. They built our idea on the prerequisites for an effective blockchain-based supply chain by interpreting and based on the developed research.[3]

This study examines blockchain technology in relation to their possible applications in logistics. Federal govt, social, and market wants to fulfil sustainability targets push us to look at how blockchain might tackle and improve procurement processes. Much of the whole rigorous assessment is determining how blockchains, a remarkable innovation in its initial stages, might transcend several possible roadblocks.

True smart contracts company's supply chain transition is still ongoing and in its initial phases; we offer upcoming research hypotheses and initiatives that might provide views into eliminating hurdles and using blockchain based for logistics operations.[4]

This study explains how a blockchain-IoT combo works 1) provides for the exchange of information and supports, resulting in the formation of a trade of goods across computers; and 2) provides us to simulates various already present, tedious activities in a hashed verified way. They also highlight several concerns to address prior to deploying a blockchain system in an IoT scenario, ranging from b2b confidentiality to the projected value of the digital goods exchanged on the system. Their assessment would be that the blockchain-IoT combo is strong and has the potential to disrupt several sectors, opening the door for innovative business strategies and unique, third-party software.[5]

This research team used a design science method to create a working prototype which has the ability to substitute an integrity café payments system predicated on a traditional, pre-paid swipe card approach.

The demonstration serves as a basis for assessing the advantages and disadvantages of distributed ledger technology when substituting an integrity structure with a security transaction system. They found that the safe and trust-free blockchain-based transaction has the potential to transform numerous present trust-based transaction technologies, but even that scaling concerns, prices, and transaction price fluctuations remain obstacles.[6]

Blockchain networks are altering traditional industries and organizational structures. Blockchain systems, which are integrated in blockchains, allow the disclosure requirements of an arrangement to be executed autonomously even with no participation of a foreign entity. Therefore, smart contracts may minimize bureaucracy and save money on solutions, increase the accuracy of company operations, and lower risks. Though blockchains have the potential to unleash a new revolutionary transformation in corporate operations, there seem to be a lot of obstacles to overcome. They begin by discussing blockchains and smart contracts. Following that, we will discuss the problems of smart contracts and also current technological breakthroughs. In addition, we evaluate common blockchain based systems and provide a classification of public blockchain apps, including some exemplary instances [7]

III. PROPOSED WORK

A. Enhancing Supply Chain Management

Distributed ledgers development has brought its way further into the realm of supply network to enable cleaner and effective operations. This has the ability to transform the method we make, sell, procure, and acquire resources. Faster sales orders execution, simpler bill computation, less use of transaction verification, and elimination of fraudulent or lost items are just a lot of small numerous benefits cryptocurrency offers the product line sector. Commitments among organizations may now be made electronic and computerized, removing the need for excessive documentation and the accompanying mistakes, slowdowns, and scam potential

dangers. Using cryptocurrency, significant economic value may be achieved by boosting accessibility, decreasing risks, and streamlining processes and overall operational control .

This is indeed a tamper-proof repository that shares several records of information throughout the whole central database, requiring an unauthorized entity to modify 51% among all occurrences in exchange to fake an addition. Technologies such as cryptocurrency to significantly reduce manufacturing costs while also increasing service quality. It acts as a go-between between the supplier and the buyer. Blockchain incorporates logistics and storage and provides such solutions to executives looking to cut operational costs. A blockchain controls all or a portion of a consumer's operational needs, that could involve shipping, stock management, inventory control, order fulfilment, or the assimilation these and many other operations. Furthermore, cryptocurrencies services will provide ten important functions such as strategic capacity, transportation knowledge, knowledge discovery, method and stockpile utilization, analytical approach, reseller process optimization, service management, prosecutable data management, best practice collaborating, and portfolio allocation dramatic drop.

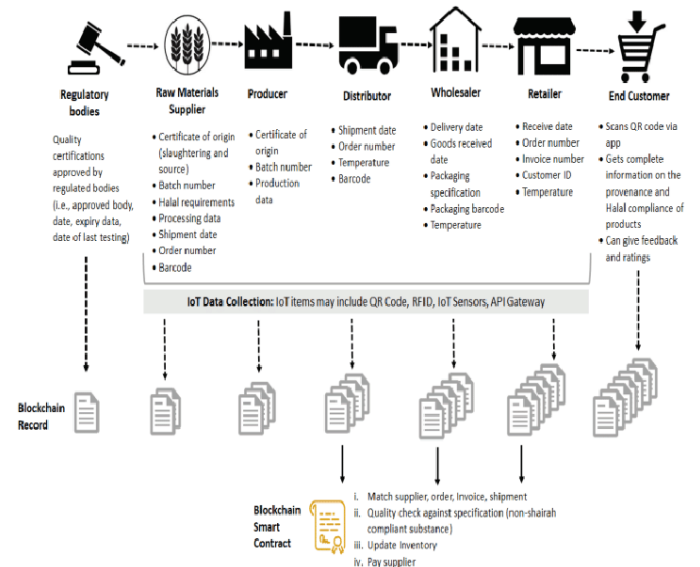


Fig. 2. Integration of Blockchain into a Supply Chain

B. Supply Chain Transparency

Transparency in the supply chain is similar to supply chain traceability in that it incorporates information that is easily available to end-users and enterprises in the supply chain. There are many levels of supply chain information sharing (also known as "visibility") inside the supply chain.

Detectability is not an alternative for business due diligence to detect any negative consequences, according to the United Nations Global Compact on Traceability. Some organizations with a thorough grasp of their supply chains and partners develop their own traceability standards, which are often used for high-value or strategic commodities. Several traceability frameworks exist for various industries with a diverse multi-stakeholder framework, such as farming and most industries, to provide reliable and stringent relevant documentation

regulations and accreditation for commodities throughout the distribution network.

C. Working of Block-Chain enabled Traceability Applications

Because of their features, blockchains seem to be well to go with tracking applications. Assets and associated documentation (such as contract of carriage or customs clearance documents) are susceptible to forgery or fraud as they move from one distribution network stakeholder to the other. To prevent this, blockchain digitalization enables the design of a digitized "token" that is connected to real products as they are produced. The token can then be authenticated by the product's ultimate owner, allowing the product's history to be traced back to its original location. End consumers have enhanced faith in the data they get since no one person or company of organizations may change the information contained within the blockchain whenever and wherever.

Due to the straight movement of most items from the point of origin to the end customer, blockchain is an ideal solution for supply chain traceability. This operational feature helps maintain anonymity because commodities and their accompanying "tokens" are seldom transferred between competitors within a blockchain. As a result, participant privacy may be preserved.

D. Coffee Bean Supply Chain

Here, we aim to solve the problem of tracking the full supply chain from the production of the coffee beans to the consumer hands. Presently, the coffee business depends heavily on faxes and email to transmit and receive transactions all over the world, leading to delayed and more error-prone documentation. We try to automate the process by emulating the functions of various participants in the coffee bean supply chain such as producers, distributors, retailers, and consumers by designing smart contracts and deploying them on the rinkeby test network. Our aim is to create a more efficient supply chain that is error free and ensures smooth working among various participants.

IV. METHODOLOGY

The coffee distribution network is the series of actions and operations that transport roasted coffee seeds from fields to marketplaces as finished espresso.

Blockchain architecture require compact resolution methods to reach optimal computational efficiency. Supply chains may be required to accommodate a high volume of transactions generated by diverse IoT devices. New compact resolution techniques are required to deal with massive volumes of dataset from IoT devices. Furthermore, a blockchain architecture must include services that enable IoT Technologies to be enrolled and authenticated prior to providing any transactional data.

The project is built using the following technologies: -

1. Ethereum – Ethereum is a decentralized platform that runs smart contracts.
2. Truffle Framework – Truffle is the most popular development framework for Ethereum which makes creating on Ethereum smoother and simpler.

3. Infura – The Infura API suite provides instant access over HTTPS and WebSockets to the Ethereum and IPFS networks

Farmers conduct ripping, sieving, refining, sorting, and classifying tasks, as well as changing crop type, temperature required, and moisture monitored throughout the operation. They are responsible for providing seeds to the distributors in packed form. Distributors are the companies that distribute cocoa all over the globe. The distributor enters the amount, the delivery location, the ship name, the ship id, the projected day and timing, and the dealer ID. Traders often purchase items from a producer, distributor, or other supplier and resale them to the common people. Consumers are the people who buy the goods from the retailer.

Refer the activity diagram below

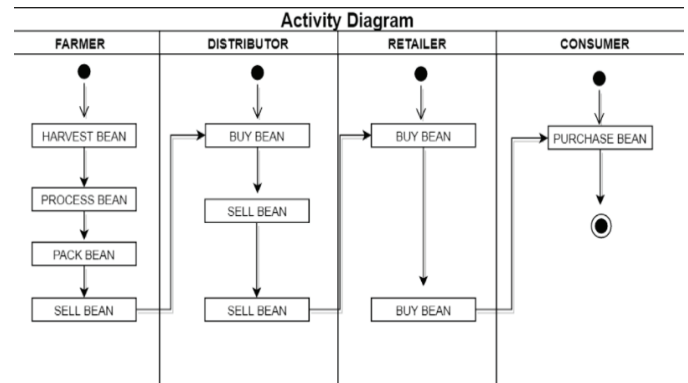


Fig. 3. Refer the activity diagram below

All the activities happening in sequential orders is shown in the below diagram.

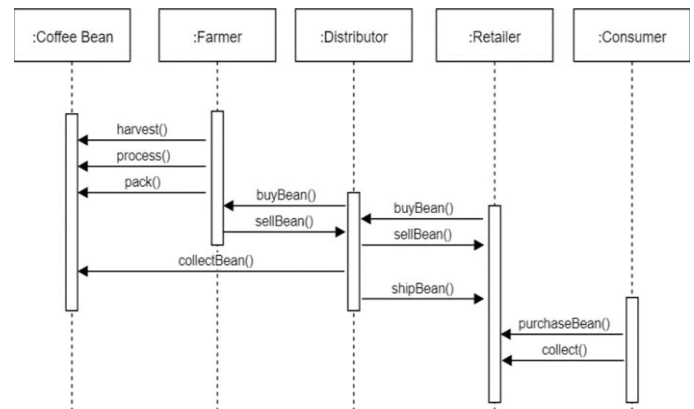


Fig. 4. Sequence diagram.

V. RESULT ANALYSIS

Here, we have created a smart contract emulating the role of a consumer in the supply chain. Similarly, contracts were made for farmers, distributors, and retailers. After creation, contracts were migrated to rinkeby test network. Refer figure below

```

1  {
2    "contractName": "ConsumerRole",
3    "abi": [
4      {
5        "inputs": [],
6        "stateMutability": "nonpayable",
7        "type": "constructor"
8      },
9      {
10       "anonymous": false,
11       "inputs": [
12         {
13           "indexed": true,
14           "internalType": "address",
15           "name": "account",
16           "type": "address"
17         }
18       ],
19       "name": "ConsumerAdded",
20       "type": "event"
21     },
22     {
23       "anonymous": false,
24       "inputs": [
25         {
26           "indexed": true,
27           "internalType": "address",
28           "name": "account",
29           "type": "address"
30         }
31       ],
32       "name": "ConsumerRemoved",
33       "type": "event"
34     },

```

Fig. 5.

The complete supply chain was deployed on the Ethereum test network. Refer figure below.

```

Deploying 'SupplyChain'
-----
> transaction hash: 0xa95e68a29fa3e7deb91317f346e0449e2a36cd7f3fe912e72d234a212971d74b
> Blocks: 1        Seconds: 12
> contract address: 0x95d4Bf8d2D67D8E11e28e9e7728f758aE159a334
> block number:    5683764
> block timestamp: 1577395787
> account:         0x4ACF5e91813f31aDF278FF22921a210972a8Cc3
> balance:         4.95651555
> gas used:        3785474
> gas price:       10 gwei
> value sent:      0 ETH
> total cost:      0.03785474 ETH

```

Fig. 6.

All the smart contracts denoting functions were tested using truffle test. Refer figure below

```

ganache-cli accounts used here...
Contract Owner: accounts[0] 0x4ACF5e91813f31aDF278FF22921a210972a8Cc3
Farmer: accounts[1] 0x9210aa648C77a3887e84e72ab2014d2b4a8a405
Distributor: accounts[2] 0xd46c0e6d278183666023fe9d620615C782d71C1
Retailer: accounts[3] 0xd806f081198978E8C9cF4dc128fChE51326Adf7F
Consumer: accounts[4] 0x8E486fF025E7C732da23e6d8d43896F215638521

Contract: Supplychain
✓ Testing smart contract function harvestItem() that allows a farmer to harvest coffee (30ms)
✓ Testing smart contract function processItem() that allows a farmer to process coffee (190ms)
✓ Testing smart contract function packItem() that allows a farmer to pack coffee (170ms)
✓ Testing smart contract function sellItem() that allows a farmer to sell coffee (230ms)
✓ Testing smart contract function buyItem() that allows a distributor to buy coffee (300ms)
✓ Testing smart contract function shipItem() that allows a distributor to ship coffee (170ms)
✓ Testing smart contract function receiveItem() that allows a retailer to mark coffee received (20ms)
✓ Testing smart contract function purchaseItem() that allows a consumer to purchase coffee (260ms)
✓ Testing smart contract function fetchItemOverview() that allows anyone to fetch item details from blockchain (40ms)
✓ Testing smart contract function fetchItemBufferTwo() that allows anyone to fetch item details from blockchain (64ms)

10 passing (2s)

```

Fig. 7.

VI. CONCLUSION

In this paper we have given the idea how blockchain can be used in the supply chain network to improve its efficiency. The blockchain will connect all components of the distribution network, making system safer and more immune, transparent, and organized. It'll also benefit all parties, especially consumers, producers, and transportations. More significantly, we aimed to describe the advantages and disadvantages of acceptance in terms of giving a basis for professionals and academics to drive their future work toward enhancing the innovation and its relevancy. In terms of real solutions, we intend to address the constraints of these kind of projects as well as their functional requirements.

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